

Fake News Detection: A Comparison of RNN and LSTM Models

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Abstract—The proliferation of fake news as a societal problem is greatly accelerated by improvements in AI, whereby false information is being created and distributed at scale. We have examined two deep learning models—Recurrent Neural Network(RNN) and Long Short-Term Memory(LSTM)—in performance to identify fakes from real news using a Fake and Real News dataset from Kaggle. Both models were trained on the dataset and tested by various metrics: accuracy, precision, recall, and F1-score. For all metrics employed, the performance of the LSTM model outperformed the RNN model. The accuracy of the LSTM model equals 97.68%, whereas the accuracy of the RNN model was 87.34%. These findings indicate that LSTM is apt for tasks requiring complicated relationship capturing along with long-term dependencies in textual data. This comparison study gives many valuable insights into the development of better fake news detection systems. Code for this project can be found: JunjieZeng1/STATS507.git

Index Terms—Fake News, classification, LSTM, RNN

I. INTRODUCTION

Fake news has been a prevailing problem in society for many years. The rapid development of AI in recent years had exacerbated the problem, resulted in significant challenges to the reliability of information sources. Accessing reliable news is no longer something we can take for granted, and detecting fake news has become increasingly essential. Various efforts were made to detect fake news. RNN and LSTM models are particularly suitable for this project because they are designed to process sequential data like text, which is essential for understanding the context and structure of news articles. In 2021, Mridha et al. reviewed various deep learning techniques, mainly hybrid, used for fake news detection and presented a structured analysis of their effectiveness [1]. Jain et al. tested LSTM model on data drawn from Kaggle and achieved a precision of 94% [2].

This project aims to explore and compare RNN and LSTM models for detecting fake news. Although various more complicated models seemed to perform well, direct comparison between simple RNN and LSTM models are not common. We strive to identify which architecture is most suitable for fake news detection and the reasons behind it. Understanding the strengths and weaknesses of each model will contribute to the development of more accurate and efficient fake news detection systems.

II. METHOD

A. Learning Problem

Fake news detection is modeled as a binary classification task where the input is a piece of news text, and the output is either “fake” or “true”.

Input: A tokenized sequence of words representing the news article

Output: A binary label (1 for “true”, 0 for “fake”)

B. Dataset

The dataset used was the Fake and Real News Dataset drawn from Kaggle [3].

Dataset separated in two files:

- Fake.csv (23502 fake news article)
- True.csv (21417 true news article)

Dataset columns:

- Title: title of news article
- Text: body text of news article
- Subject: subject of news article
- Date: publish date of news article

C. Model Formulation

Two deep learning models were implemented:

Recurrent Neural Network (RNN) with three layers, embedding layer, RNN layer, and fully connected layer

Long Short-Term Memory (LSTM) with three layers, embedding layer, LSTM layer, and fully connected layer

D. Methodologies

The dataset was split into 80% training set and 20% testing set. Both RNN and LSTM models were trained for 5 epochs using a batch size of 64, along with Adam optimizer with a learning rate of 0.001. The loss function used for training was CrossEntropyLoss, which is appropriate for classification tasks. The performance of the models was evaluated on testing set via various evaluation metrics: accuracy, recall, precision, and F1 score.

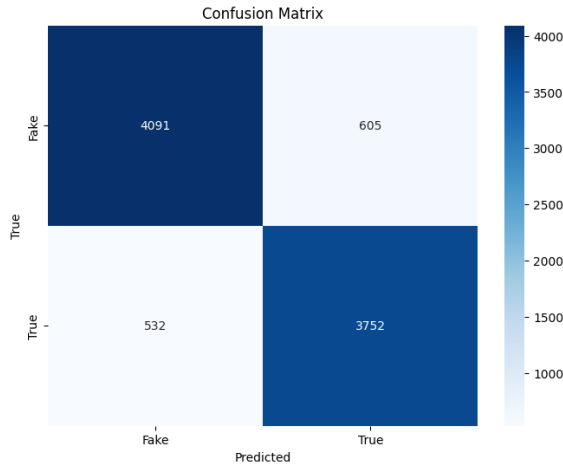


Fig. 1. Confusion Matrix for RNN Model

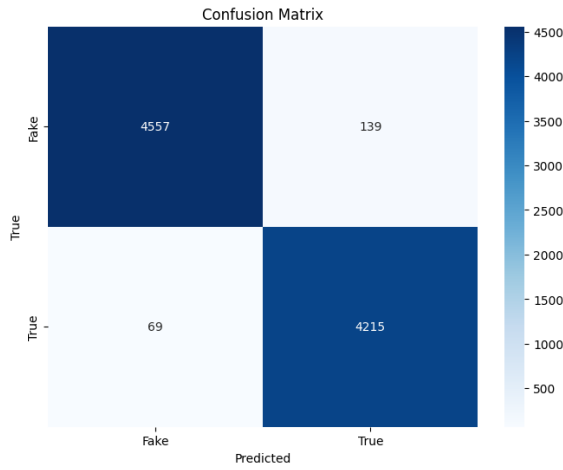


Fig. 2. Confusion Matrix for LSTM Model

III. RESULTS

A. Data Preprocessing and Model Setup

Two datasets were merged and labeled with True and False. Title and Text columns were combined into one Text column. Subject and Date columns were not used in this project. After the combined dataset was obtained, it was preprocessed through following steps:

- Removed non-alphabetic characters
- Lowercased every word
- Splited into 80% train set, 20% test set
- Tokenization
- Padded sequences with a maximum length of 1000 words

After data preprocessing, RNN and LSTM models were developed. Both models were developed similarly for a fair comparison. Input size was set to be the vocabulary size (which is up to 1000). Hidden dimension was set to be 64. Output dimension was 2 (1 for true, 0 for fake). Both models were trained for 5 epochs and their performance were evaluated.

B. Interpretation of the Results

TABLE I
PERFORMANCE OF RNN AND LSTM MODELS ON TEST SET

Model	Performance Criteria			
	Accuracy%	Precision	Recall	F1 Score
RNN	87.34	0.87	0.87	0.87
LSTM	97.68	0.98	0.98	0.98

Confusion matrices were visualized for the correct and incorrect predictions made by the RNN model (figure 1) and the LSTM model (figure 2) on the test set. Table I provides the performance of the RNN and LSTM models on the test set for four metrics: Accuracy, Precision, Recall, and F1 Score. From the table, it is clear that the LSTM model outperforms the RNN model across all metrics. The LSTM model achieved accuracy of 97.68%, whereas the RNN model had an accuracy of 87.34%. The large difference in accuracy(10%) and F1 Score(0.11) indicates that LSTM captures more complex relationships in the data, while RNN is less capable of identifying complex relationship. This is reasonable since structurally, LSTM networks are designed to handle long-term dependencies in sequential data better than traditional RNNs, making them more suited for text-based tasks like fake news detection.

IV. CONCLUSION

In this paper, we compared the RNN and LSTM models for fake news detection models using the Fake and Real News Dataset. Data were preprocessed and the RNN and LSTM models were trained and evaluated. The results indicate that the LSTM model significantly outperforms the RNN model, achieving higher accuracy, precision, recall, and F1-scores. We conclude the LSTM model is better at fake news detection tasks and we attribute this to its ability to capture long-term dependencies and complex relationships in textual data. These findings provide a foundation for future research.

Further research can investigate the performance of more advanced models such as Bidirectional Encoder Representations from Transformers (BERT) and other transformer-based architectures that are known for their contextual understanding and could provide even higher accuracy in fake news detection.

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